



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: ECE / EEE F434
Course Title	: Digital Signal Processing
Instructor-in-Charge	: Sandeep Joshi
Instructor(s)	: Sandeep Joshi
Tutorial Instructors	: Sharda Tripathi and Sandeep Joshi
Practical Instructors	: Rahul Runthala (RS), Kanwardeep Singh Gahlot (RS), Aniket Kumar (RS), Sudhir Kumar (RS)

1. Course Description:

Introduction; design of analog filters; design of digital filters (IIR and FIR); structures for the realization of digital filters; random signals and random processes; linear estimation and prediction; Wiener filters; DSP processor architecture; DSP algorithms for different applications.

2. Scope and Objective of the Course:

- To provide a comprehensive knowledge of DSP's fundamental theory, concepts, and applications.
- To be able to design and simulate practical DSP-based systems, finding applications in various fields, namely, communication engineering, audio, video, and image signal processing.

3. Course Learning Outcomes:

- Use Fourier analysis (DFT and FFT) for efficient frequency domain analysis of digital signals.
- Design and implement digital IIR and FIR filters for signal processing and communication system applications.
- Simulate and analyze DSP algorithms using tools like MATLAB and CCS for practical signal processing applications.

4. Text Books:

- S. K. Mitra, Digital Signal Processing, McGraw-Hill Higher Education, 3rd ed., 2005
- John G. Proakis and D.G. Manolakis, Digital Signal Processing: Principle, Algorithms and Applications, Pearson Education, 4th ed., 2007.

Reference Books:

- A.V. Oppenheim and Schafer, Discrete Time Signal Processing, 3rd ed., Pearson Education, 2013.
- J. R. Johnson, Introduction to Digital Signal Processing, Prentice Hall, 1992.
- Richard G. Lyons, Understanding Digital Signal Processing, Pearson Education, 2nd ed., 2004.

- Course will be conducted through weekly lecture sessions (3hrs/week), tutorial sessions (1hr/week), and lab sessions (2hrs/week). The lecture sessions will make extensive use of Blackboard and conceptual understanding. The tutorial sessions will be focused on problem solving and lab sessions will cover simulation-based and hardware-based exercises over a period of 14 weeks covering the semester.



6. Course Plan:

Module	Lecture Session	Coverage	Ref.	Learning outcomes
Introduction to digital signal processing (DSP)	1-4	Introduction to DSP and its applications; review of signal theory- CT, DT signals and systems	Ch-1 (T1, T2) and class notes	What is DSP, and the need for DSP?
	5-6	LTI systems, FIR, IIR; system stability		
DFT	7-9	DTFT, Discrete Fourier Transform (DFT): properties, applications, circular convolution	Ch-7, 8 (T2), T1 and class notes	DFT applications, limitations, and need for efficient computations. FFT techniques
	10-16	Efficient computation of DFT- Fast Fourier Transform (FFT); FFT algorithms, FFT applications: OFDM		
Frequency domain analysis	17-19	Review of Z-transforms, random signals, and processes	Ch-5, 6 (T2) and class notes	Frequency domain analysis, processes and advantages
	20-21	Frequency response of LTI systems; Sampling and reconstruction of signals: ADC, DAC		
Digital filters	22-25	Implementation of DT systems: FIR, IIR, number representation, quantization	Ch-9, 10 (T2), and class notes	Digital filter design techniques and practical considerations for system realization
	26-29	Analog filter design: Butterworth, Chebyshev, Elliptical		
	30-37	Digital filter design: FIR, IIR; transformation-IIT, BLT; windowing, filter structures: Lattice		
DSP Processors [Majorly covered in Lab]	38-39	DSP arithmetic, an overview of programmable DSPs, DSP processor architecture: TMS DSP processor	Class notes, TI Manual	Overview of DSP processors
Advanced topics and Applications	40-42	Multirate DSP; Adaptive filters, linear estimation, and prediction; AI applications in DSP	Ch-11 (T2), class notes, and Research papers	New trends in DSP and various applications

Lab

DSP lab experiments focus on the understanding of the system design and simulation, helping in appreciating and complementing the concepts developed in the lectures. The lab experiments include simulation-based experiments and hardware experiments. The simulation experiments will be based on MATLAB, and the hardware experiments will introduce the students to DSP processor architecture and Code Composer Studio (CCS). The lab experiments list is as follows.



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- Introduction to DT signal operations and familiarization with MATLAB.
- DT Signal operations: Convolution, correlation, and applications of the Toeplitz matrix.
- Sampling and DTFT
- Circular convolution
- DFT and FFT
- Multirate DSP
- Filter design: IIR and FIR
- Experiments on the DSP Kit and introduction to CCS, DSP architectures
- Introduction to FPGA boards for DSP applications [planned]
- Lab Project: Research paper implementation and extension

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Marks (Total 200)	Date & Time	Nature of component
Mid-Semester Test	90 Min.	30	60	07/10/2026 AN1	Closed Book
Lab (Regularity and Project)	--	15	30	To be announced in class	Open Book
Quiz (s)	--	15	30	To be announced in class	Open Book
Comprehensive Examination	180 Min.	40	80	10/12/2026 FN	Closed Book/Open Book

8. Attendance Policy: Student is expected to attend all lectures/tuts/lab and appear for all evaluative components.

9. Chamber Consultation Hour: To be announced in the lecture.

10. Notices: All announcements will be made in the lectures or posted on Nalanda LMS.

11. Make-up Policy: No make-up will be given for quizzes; however, for other components, make-up will be given **ONLY** in cases of sickness (hospitalization). In such cases, students must produce sufficient proof and must pre-inform the Instructor-in-charge.

12. Minimum performance requirement for a valid grade: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

13. Note (if any):

- Programmable calculators are not allowed.
- Strict action, including awarding of zero marks, will be taken in case of plagiarism of the lab codes and/or lab reports. All submissions will be checked by a plagiarism detection tool.
- Regular attendance is strongly recommended for the lab sessions.

Sandeep Joshi
Instructor-in-charge
Course No. EEE F434 / ECE F434